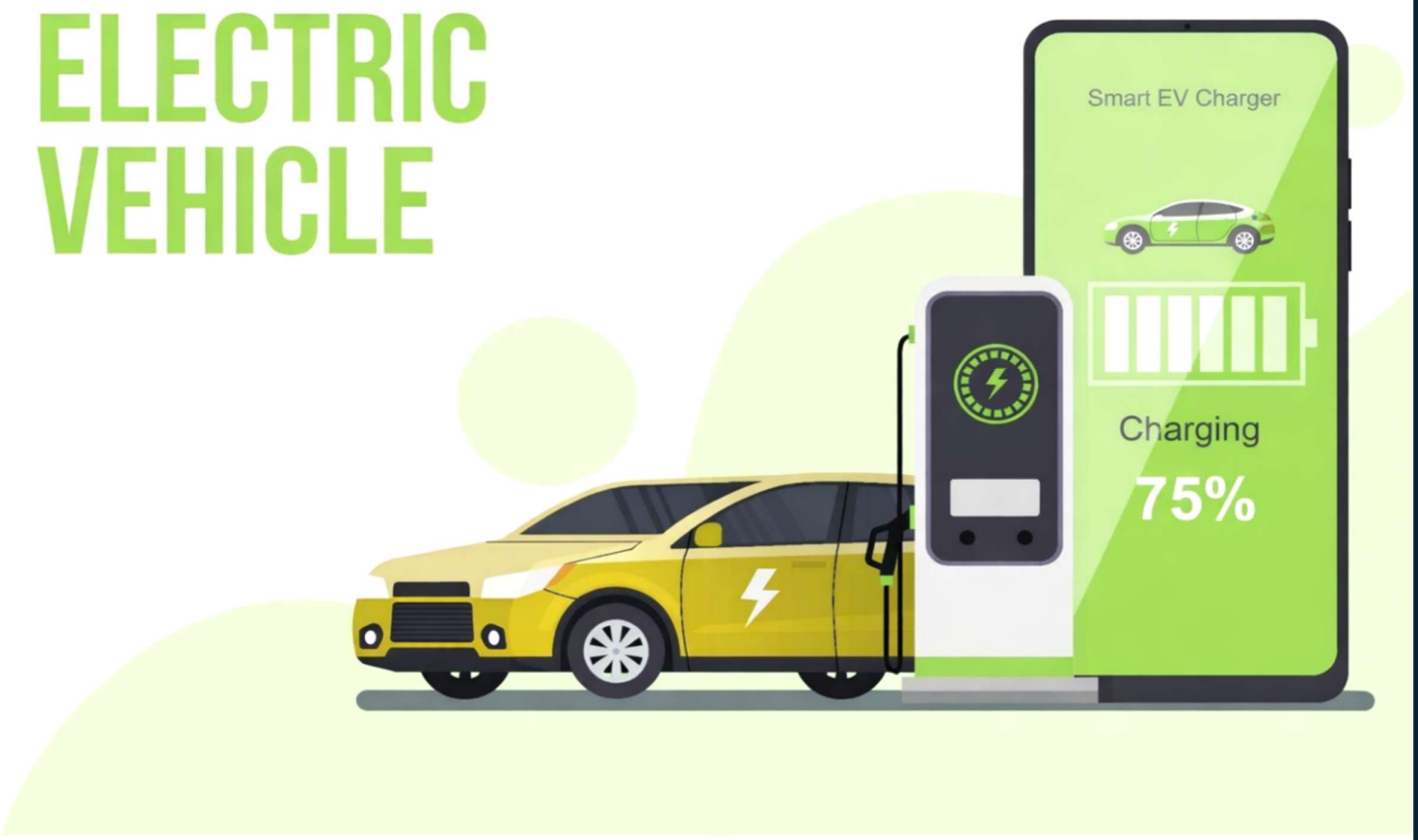
EXPLORATORY DATA ANALYSIS

CAPESTONE PROJECT



INTRODUCTION

Electric Vehicles (EVs) are becoming an important part of the modern transportation system due to their eco-friendly nature and reduced dependence on fossil fuels. With the rapid growth of EV adoption, the insurance industry is also evolving to understand customer behaviour, risk factors, and policy preferences related to vehicle ownership.

This project focuses on analysing a dataset that contains customer information such as age, gender, driving license status, vehicle age, insurance history, and annual premium. The dataset also includes whether the customer has previously been insured and whether their vehicle has been damaged, which are key factors in determining insurance demand.

The main objective of this analysis is to identify patterns and insights that influence a customer's decision to purchase vehicle insurance (Response). By studying variables like vehicle age, premium amount, and past insurance behaviour, we can better understand customer segments and predict future insurance trends, especially in the context of electric vehicles.

This analysis can help insurance companies design better policies, improve customer targeting, and support the growing EV ecosystem by offering more tailored and efficient insurance solutions.

OBJECTIVES

1.Customer Analysis

This objective focuses on analysing customer details such as age, gender, and driving license status to understand how these factors affect their decision to purchase insurance. It helps identify which group of customers is more likely to buy EV insurance policies.

2.Vehicle Factors

This objective studies how vehicle-related aspects like vehicle age and vehicle damage influence insurance demand. It helps in understanding how diverse types of vehicles, including electric vehicles, impact customer decisions.

3.Insurance History

This objective examines the effect of previous insurance status on customer behaviour. It helps identify whether customers who were previously insured are more likely to purchase insurance again.

4.Premium Impact

This objective analyses how the annual premium amount affects the customer's decision to buy insurance. It helps in understanding price sensitivity and setting suitable pricing strategies.

5.Purchase Prediction

This objective aims to identify patterns in the data to predict whether a customer will purchase insurance or not. It helps companies improve targeting and decision-making for EV insurance policies.

IMPORTANT PYTHON LIBRARIES

1. Pandas (import pandas as pd)

Pandas is the main library used in this project. It helps in loading the dataset and handling structured data like customer details, vehicle age, premium, and response. It is used for data cleaning, filtering, grouping, and analysis.

2. NumPy (import numpy as np)

NumPy is used for numerical calculations such as averages, percentages, and data transformations. It helps in analysing values like annual premium, age distribution, and response rate.

3. Matplotlib (import matplotlib.pyplot as plt)

Matplotlib is used to create basic visualizations such as bar charts, histograms, and scatter plots. It helps in understanding relationships like age vs response or premium vs response.

4. Seaborn (import seaborn as sns)

Seaborn is used for advanced visualization. It helps in creating heatmaps and comparison plots to analyse relationships between variables like vehicle damage, insurance history, and response.

DATA EXPLORATION

Loading Dataset

```python
#Load DataSet
import pandas as pd
df = pd.read_csv("C:/Users/yadav/Downloads/Vehicle_Insurance.csv")
```

I have loaded the Vehicle Insurance dataset into the Python environment using the Pandas library. The **pd.read_csv()** function is used to import the dataset from the specified file path and store it in a Data Frame named **df**, which serves as the primary data structure for further analysis. The dataset contains important columns such as **Gender, Age, Driving_License, Region_Code, Previously_Insured, Vehicle_Age, Vehicle_Damage, Annual_Premium, Policy_Sales_Channel, Vintage, and Response**, which provide detailed information about customers and their insurance behaviour.

Pandas enables efficient handling of structured data and provides flexible tools for data cleaning, manipulation, and exploration. This allows easy analysis of customer demographics, vehicle details, premium patterns, and their impact on insurance purchase decisions. This step forms the foundation of the project, ensuring that the dataset is successfully loaded and ready for exploratory data analysis (EDA).

Understanding Dataset Information

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 381109 entries, 0 to 381108
Data columns (total 12 columns):
 #   Column                Non-Null Count   Dtype
---  ------                --------------   -----
 0   id                    381109 non-null  int64
 1   Gender                381109 non-null  object
 2   Age                   381109 non-null  int64
 3   Driving_License       381109 non-null  int64
 4   Region_Code           381109 non-null  float64
 5   Previously_Insured    381109 non-null  int64
 6   Vehicle_Age           381109 non-null  object
 7   Vehicle_Damage        381109 non-null  object
 8   Annual_Premium        381109 non-null  float64
 9   Policy_Sales_Channel  381109 non-null  float64
 10  Vintage               381109 non-null  int64
 11  Response              381109 non-null  int64
dtypes: float64(3), int64(6), object(3)
memory usage: 34.9+ MB
```

The dataset contains **381,109 rows and 12 columns**, which shows it is a large dataset. All columns have **non-null values**, so there are **no missing values** present.

I also observed that the dataset includes different data types such as **int, float, and object**. Columns like **Age, Driving_License, and Response are numerical, while Gender, Vehicle_Age, and Vehicle_Damage are categorical**.

This helped me understand that the dataset is clean and well-structured, and ready for further analysis.

DATA EXPLORATION

Understanding Dataset Information

```
df.head()
```

	id	Gender	Age	Driving_License	Region_Code	Previously_Insured	Vehicle_Age	Vehicle_Damage	Annual_Premium	Policy_Sales_Channel	Vintage	Response
0	1	Male	44	1	28.0	0	> 2 Years	Yes	40454.0	26.0	217	1
1	2	Male	76	1	3.0	0	1-2 Year	No	33536.0	26.0	183	0
2	3	Male	47	1	28.0	0	> 2 Years	Yes	38294.0	26.0	27	1
3	4	Male	21	1	11.0	1	< 1 Year	No	28619.0	152.0	203	0
4	5	Female	29	1	41.0	1	< 1 Year	No	27496.0	152.0	39	0

I used the **df.head()** function to view the first 5 rows of the dataset. This helped me quickly understand how the data looks and what kind of values are present in each column.

I found that each row represents a **customer record** with details like *Gender, Age, Vehicle_Age, Annual_Premium,* and *Response*. It also helped me verify that the dataset is properly loaded and the columns are correctly aligned.

This step gave me a quick overview of the dataset before starting further analysis.

Checking Dataset Size

```
df.shape
```

```
(381109, 12)
```

I used the **df.shape** function to check the size of the dataset. I found that the dataset contains **381,109 rows and 12 columns**.

This means there are **381,109 customer records** and **12 different features** in the dataset. It helped me understand the overall size of the data and confirmed that it is large enough for analysis.

DATA EXPLORATION

Types of Information in Dataset

```python
print(df.dtypes)

numerical_cols = ['Age', 'Region_Code', 'Annual_Premium',
                  'Policy_Sales_Channel', 'Vintage']
print("Numerical Columns:", numerical_cols)

categorical_cols = ['Gender', 'Vehicle_Age', 'Vehicle_Damage']
print("Categorical Columns:", categorical_cols)

binary_cols = ['Driving_License', 'Previously_Insured', 'Response']
print("Binary Columns:", binary_cols)
```

```
id                        int64
Gender                   object
Age                       int64
Driving_License           int64
Region_Code             float64
Previously_Insured        int64
Vehicle_Age              object
Vehicle_Damage           object
Annual_Premium          float64
Policy_Sales_Channel    float64
Vintage                   int64
Response                  int64
dtype: object
Numerical Columns: ['Age', 'Region_Code', 'Annual_Premium', 'Policy_Sales_Channel', 'Vintage']
Categorical Columns: ['Gender', 'Vehicle_Age', 'Vehicle_Damage']
Binary Columns: ['Driving_License', 'Previously_Insured', 'Response']
```

I used the code to identify different types of data present in the dataset. I found that the dataset contains **numerical, categorical, and binary data**.

The **numerical columns** include *Age, Region_Code, Annual_Premium, Policy_Sales_Channel,* and *Vintage*, which represent measurable values. The **categorical columns** include *Gender, Vehicle_Age,* and *Vehicle_Damage*, which represent different categories.

I also found **binary columns** like *Driving_License, Previously_Insured,* and *Response*, which contain values 0 and 1.

This helped me clearly understand the type of each column and made it easier to perform further analysis.

DATA EXPLORATION

Handling Missing Values and Outliers

```python
print(df.isnull().sum())

num_cols = ['Age', 'Region_Code', 'Annual_Premium',
            'Policy_Sales_Channel', 'Vintage']

for col in num_cols:
    df[col].fillna(df[col].mean())

cat_cols = ['Gender', 'Vehicle_Age', 'Vehicle_Damage']

for col in cat_cols:
    df[col].fillna(df[col].mode()[0])

Q1 = df['Annual_Premium'].quantile(0.25)
Q3 = df['Annual_Premium'].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

df = df[(df['Annual_Premium'] >= lower_bound) &
        (df['Annual_Premium'] <= upper_bound)]

print("Outliers removed successfully")
```

```
id                      0
Gender                  0
Age                     0
Driving_License         0
Region_Code             0
Previously_Insured      0
Vehicle_Age             0
Vehicle_Damage          0
Annual_Premium          0
Policy_Sales_Channel    0
Vintage                 0
Response                0
dtype: int64
Outliers removed successfully
```

I first checked for missing values using **df.isnull().sum()** and found that there were **no missing values** in the dataset. However, I applied methods to handle missing values if they were present, such as filling numerical columns with the **mean** and categorical columns with the **mode**.

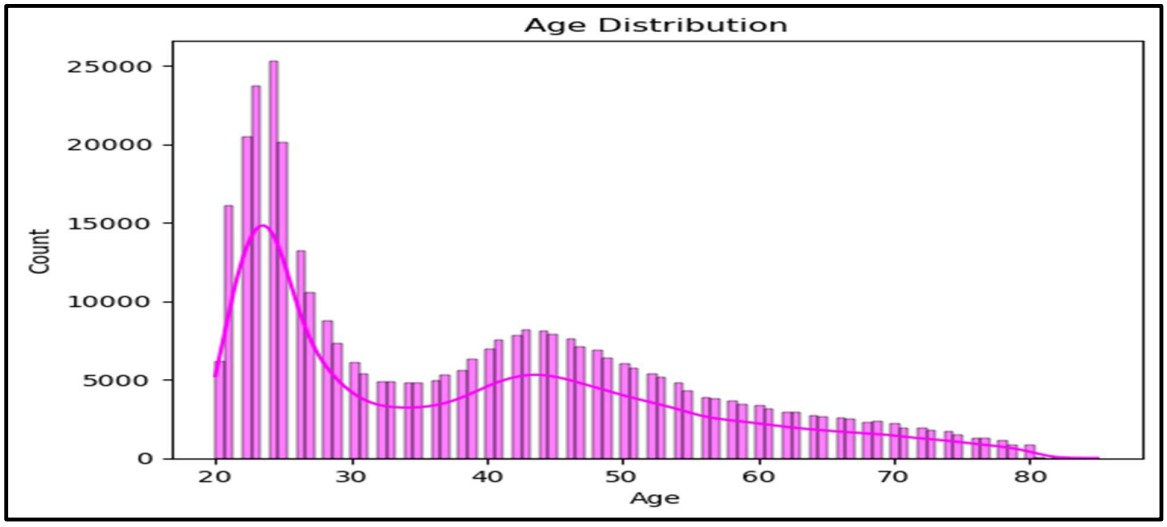
Next, I handled outliers in the dataset using the **IQR (Interquartile Range) method**, especially for the *Annual_Premium* column. I calculated the lower and upper limits and removed values that fall outside this range.

This step helped me ensure that the dataset is **clean, consistent, and free from extreme values**, making it more suitable for accurate analysis and model building.

DATA VISUALISATION

Age Distribution

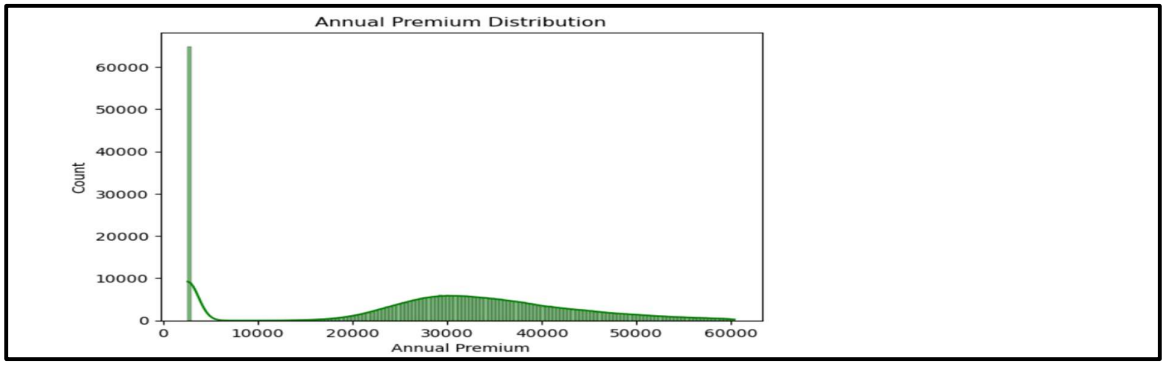
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure()
sns.histplot(df['Age'], kde=True, color = 'magenta' )
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



I plotted a histogram for Age to understand how customers are distributed across different age groups. I found that most customers fall within a specific age range, which helps in identifying the target customer segment.

Annual Premium Distribution

```
plt.figure()
sns.histplot(df['Annual_Premium'], kde=True, color = 'green')
plt.title("Annual Premium Distribution")
plt.xlabel("Annual Premium")
plt.ylabel("Count")
plt.show()
```

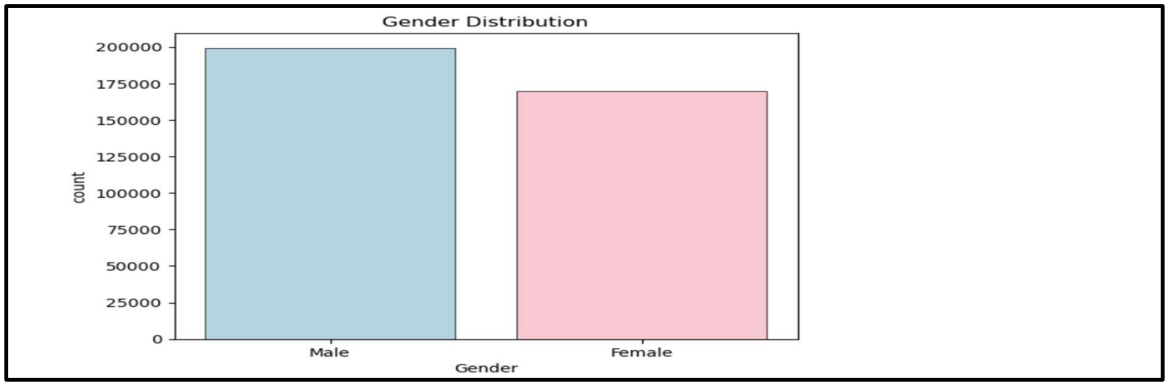


I used a histogram to analyse the distribution of Annual Premium. I found that most customers pay within a certain premium range, while very high values are less frequent, indicating possible outliers.

DATA VISUALISATION

Gender Distribution

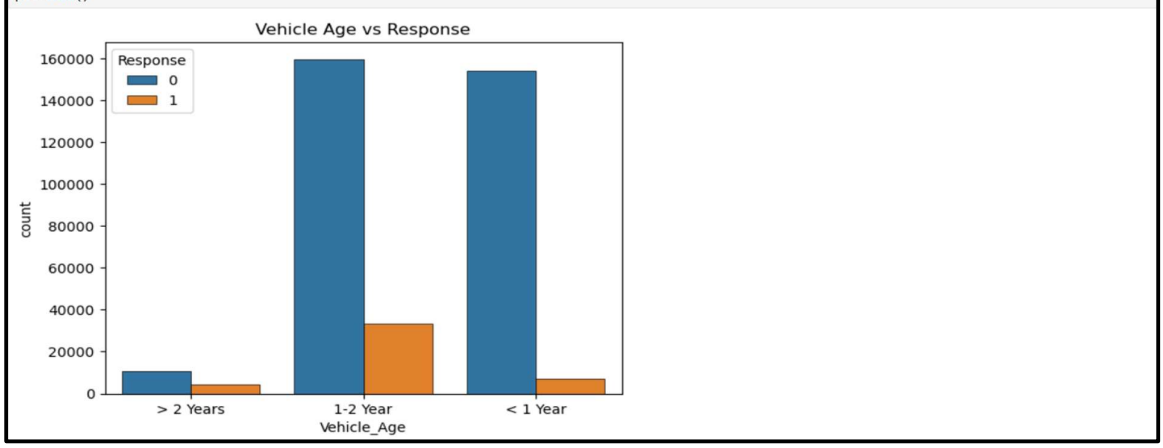
```
plt.figure()
colors = ['lightblue', 'pink']

sns.countplot(x='Gender', hue = 'Gender', data=df,
              palette=colors, edgecolor='black', linewidth=0.5, legend = False)

plt.title("Gender Distribution")
plt.show()
```



I created a count plot for Gender to see the number of male and female customers. I found the proportion of each gender, which helps in understanding customer demographics.

Vehicle Age vs Response

```
plt.figure()
sns.countplot(x='Vehicle_Age', hue='Response', data=df, edgecolor = 'black', linewidth = 0.5)
plt.title("Vehicle Age vs Response")
plt.show()
```

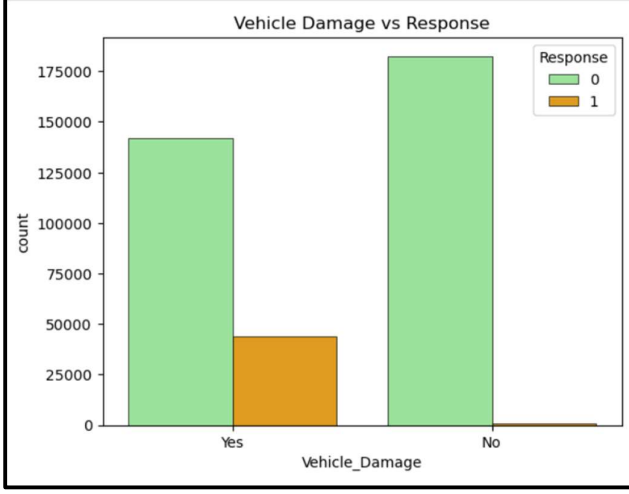


I plotted Vehicle Age against Response to analyse how vehicle age affects insurance purchase. I found that customers with older vehicles are more likely to buy insurance compared to newer vehicles.

DATA VISUALISATION

Vehicle Damage vs Response

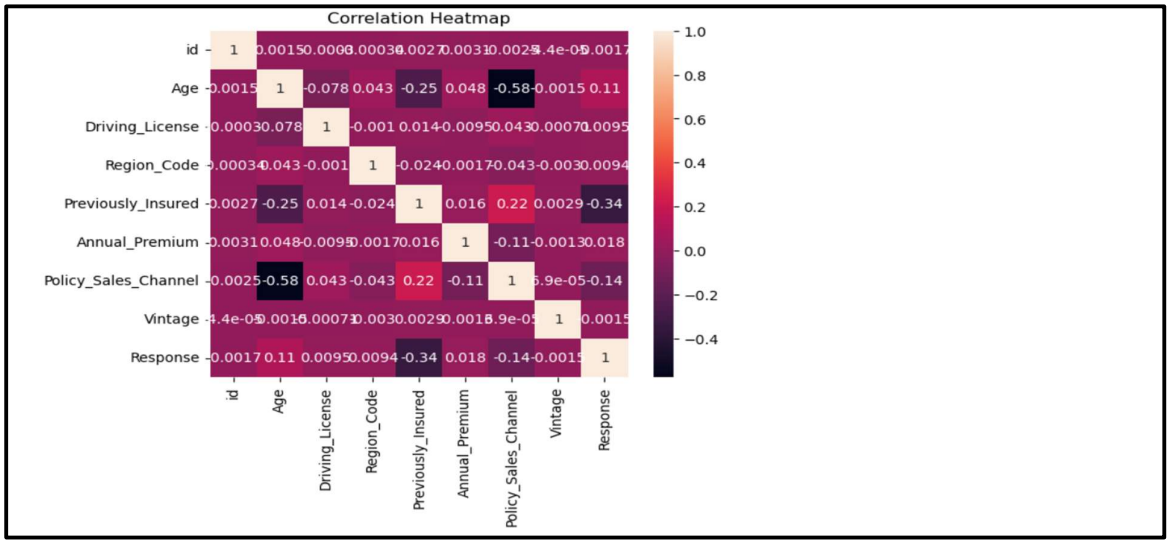
```
plt.figure()
colors = ['lightGreen','Orange']
sns.countplot(x='Vehicle_Damage', hue='Response', palette = colors, data=df, edgecolor = 'black', linewidth = 0.5)
plt.title("Vehicle Damage vs Response")
plt.show()
```



I created a count plot to compare Vehicle Damage with Response. I found that customers whose vehicles were previously damaged are more likely to purchase insurance.

Correlation Heatmap

```
plt.figure()
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```

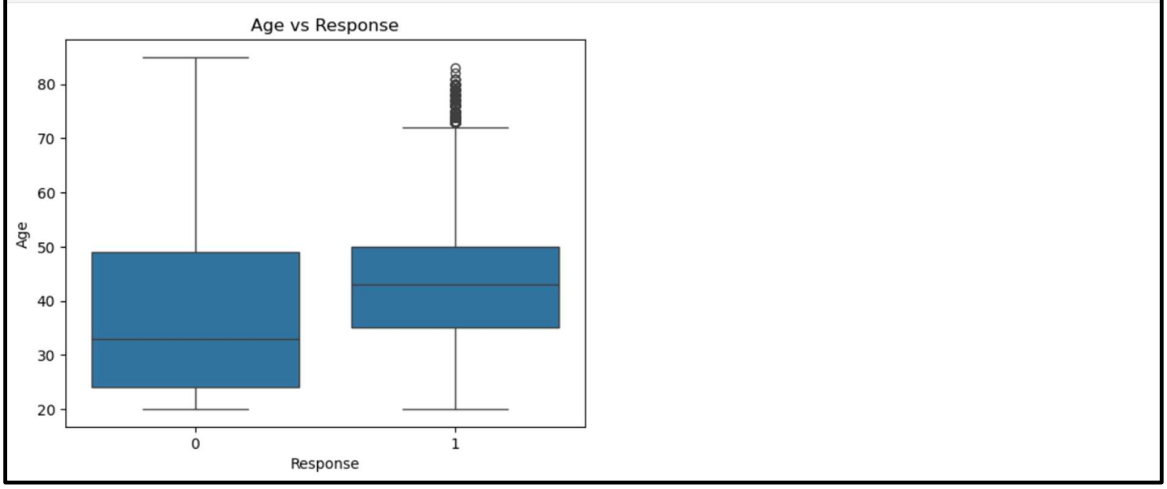


I used a heatmap to analyse the correlation between numerical variables. I found that some variables have positive or negative relationships, which helps in understanding how features are related to each other.

DATA ANALYSIS

Age vs Response

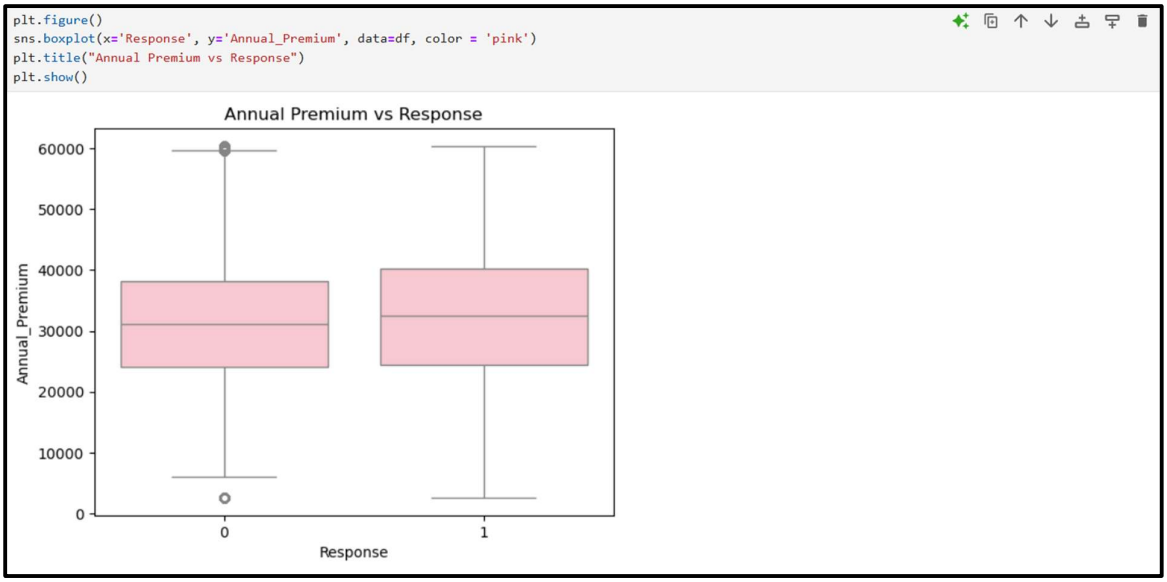
```
plt.figure()
sns.boxplot(x='Response', y='Age', data=df)
plt.title("Age vs Response")
plt.show()
```



I used a boxplot to compare Age with Response. I found that age groups show variation in insurance purchase, helping identify which age group is more likely to buy insurance.

Annual Premium vs Response

```
plt.figure()
sns.boxplot(x='Response', y='Annual_Premium', data=df, color = 'pink')
plt.title("Annual Premium vs Response")
plt.show()
```

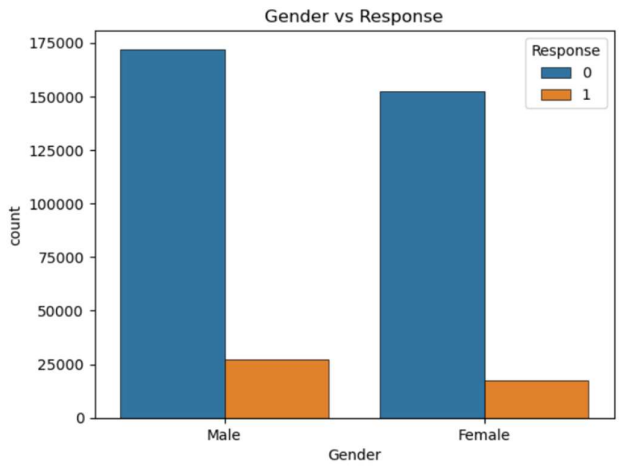


I plotted Annual Premium against Response to understand price impact. I found that premium values influence customer decisions, and extreme values may affect response.

DATA ANALYSIS

Gender vs Response

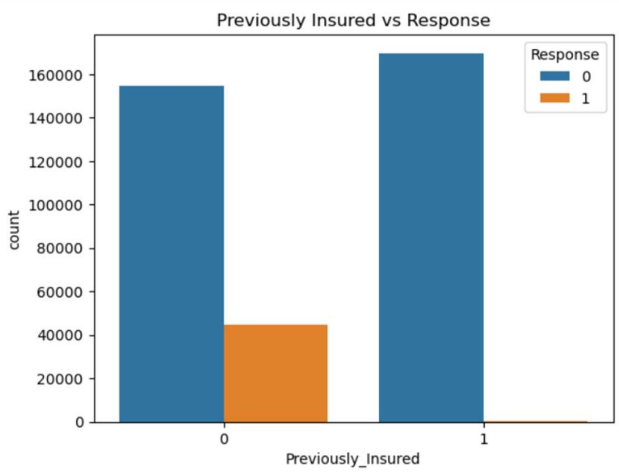
```
plt.figure()
sns.countplot(x='Gender', hue='Response', data=df, edgecolor = 'black', linewidth = 0.5)
plt.title("Gender vs Response")
plt.show()
```



I created a count plot for Gender with Response. I found how male and female customers differ in purchasing insurance.

Previously Insured vs Response

```
plt.figure()
sns.countplot(x='Previously_Insured', hue='Response', data=df)
plt.title("Previously Insured vs Response")
plt.show()
```

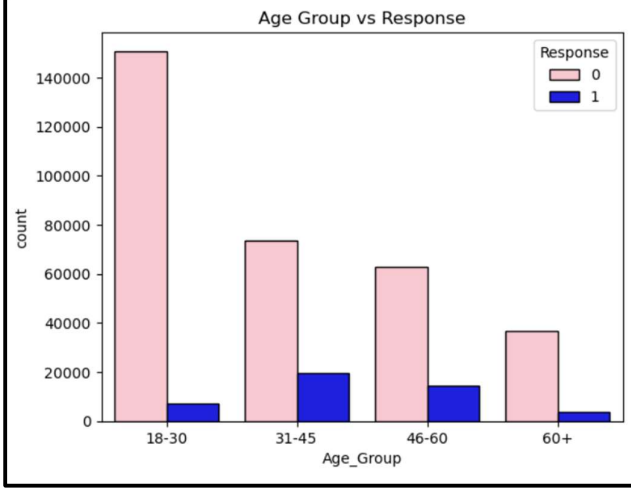


I compared Previously Insured with Response. I found that customers who were not previously insured are more likely to buy insurance.

DATA ANALYSIS

Age Group vs Response

```
plt.figure()

sns.countplot(x='Age_Group', hue='Response', data=df,
              palette=['pink', 'blue'], edgecolor='black')

plt.title("Age Group vs Response")
plt.show()
```



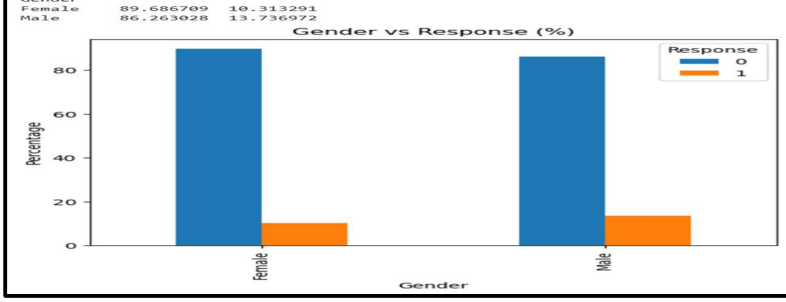
I used a count plot to analyse Age Group with Response and applied different colors to distinguish between customers who purchased insurance and those who did not.

I found that using different colours makes the comparison clearer and helps in easily identifying which age group has a higher response rate.

Gender vs Response%

```
gender_response = pd.crosstab(df['Gender'], df['Response'], normalize='index') * 100
print(gender_response)

gender_response.plot(kind='bar')
plt.title("Gender vs Response (%)")
plt.ylabel("Percentage")
plt.show()
```

```
Response            0          1
Gender
Female      89.686709  10.313291
Male        86.263028  13.736972
```



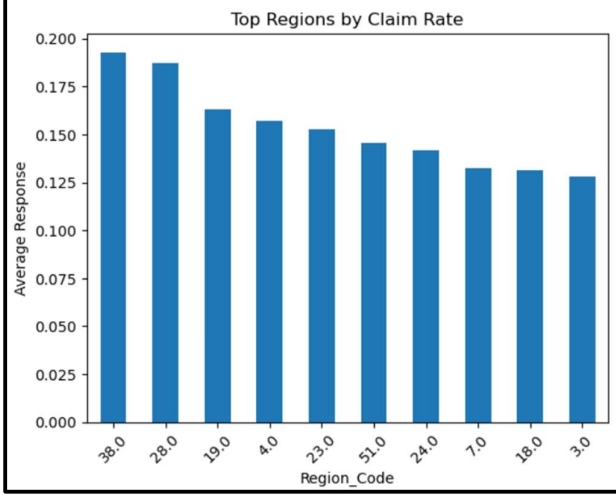
I analysed the count plot to compare Gender with Response. I found that although the number of male customers is higher, the proportion of customers who purchased insurance is almost similar for both genders.

This shows that **gender has a minimal impact** on insurance purchase decisions.

DATA ANALYSIS

Average Response by Region

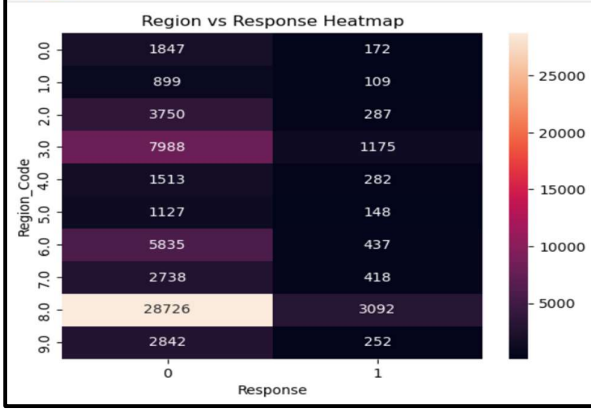
```
region_avg = df.groupby('Region_Code')['Response'].mean().sort_values(ascending=False)

plt.figure()
region_avg.head(10).plot(kind='bar')
plt.title("Top Regions by Claim Rate")
plt.ylabel("Average Response")
plt.xticks(rotation=45)
plt.show()
```



I analysed the average response rate for each region using group by. I found that some regions have a higher average response value, which means customers in those regions are more likely to purchase insurance. This helped me identify high-demand regions.

Heatmap (Region vs Response)

```
region_heat = pd.crosstab(df['Region_Code'], df['Response'])

plt.figure()
sns.heatmap(region_heat.head(10), annot=True, fmt='d')
plt.title("Region vs Response Heatmap")
plt.show()
```

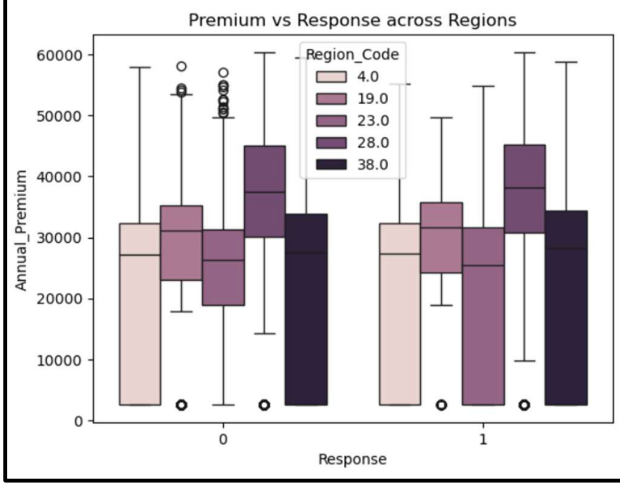


I analysed the heatmap to compare the number of customers who responded (0 and 1) across different regions. I found clear differences in response distribution, showing that some regions have more positive responses than others.

DATA ANALYSIS

Premium vs Response across Regions

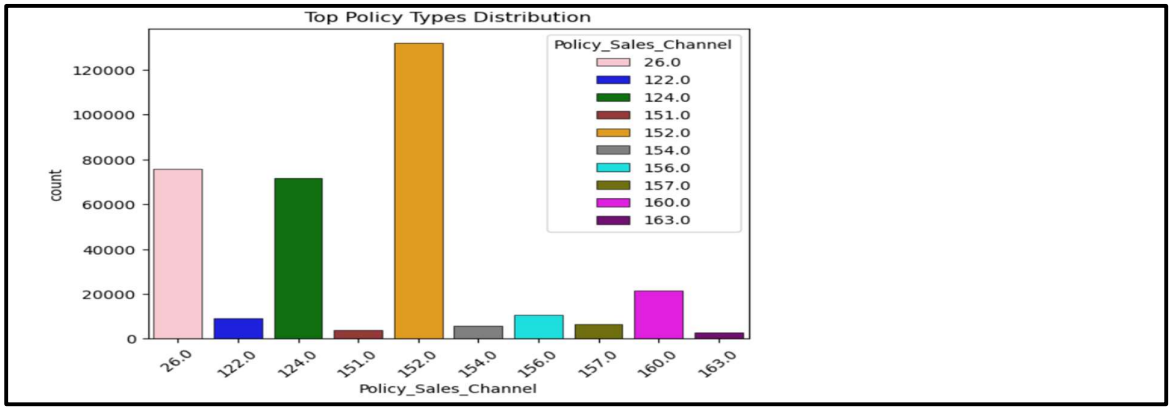
```
plt.figure()
sns.boxplot(x='Response', y='Annual_Premium', hue='Region_Code',
            data=df[df['Region_Code'].isin(region_avg.head(5).index)])
plt.title("Premium vs Response across Regions")
plt.show()
```



I analysed the boxplot to understand how premium varies across regions and response. I found that premium distribution is different in each region, and in some regions higher premiums are associated with higher response rates.

Distribution of Policy Types

```
top_policy = df['Policy_Sales_Channel'].value_counts().head(10).index

plt.figure()
colors = ['pink','blue','green','brown','orange','grey','cyan','olive','magenta','purple']
sns.countplot(x='Policy_Sales_Channel', hue='Policy_Sales_Channel',data=df[df['Policy_Sales_Channel'].isin(top_policy)],
              palette = colors, edgecolor = 'black', linewidth = 0.5)
plt.title("Top Policy Types Distribution")
plt.xticks(rotation=45)
plt.show()
```

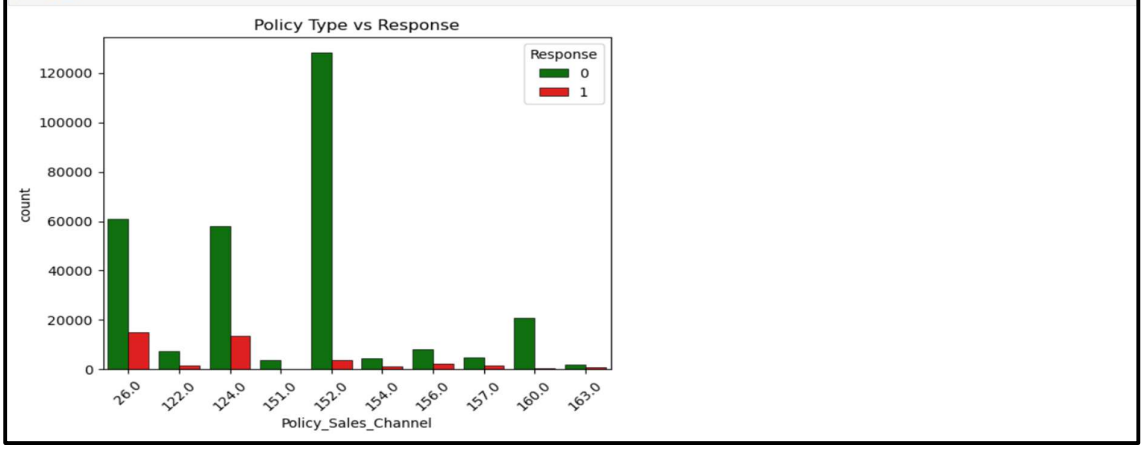


I analysed the distribution of Policy Sales Channel to understand which policy types are most common. I found that only a few policy channels have a high number of customers, while others have very low storage.

DATA ANALYSIS

Policy Type vs Response

```
plt.figure()
sns.countplot(x='Policy_Sales_Channel', hue='Response',
              data=df[df['Policy_Sales_Channel'].isin(top_policy)], palette = ['green', 'red'],edgecolor = 'black', linewidth = 0.5)
plt.title("Policy Type vs Response")
plt.xticks(rotation=45)
plt.show()
```



I analysed the relationship between policy type and Response using a count plot. I found that some policy types have a higher number of customers who purchased insurance, indicating better performance of certain channels.

Percentage Analysis

```
policy_response = pd.crosstab(df['Policy_Sales_Channel'], df['Response'], normalize='index') * 100
print(policy_response.loc[top_policy])
```

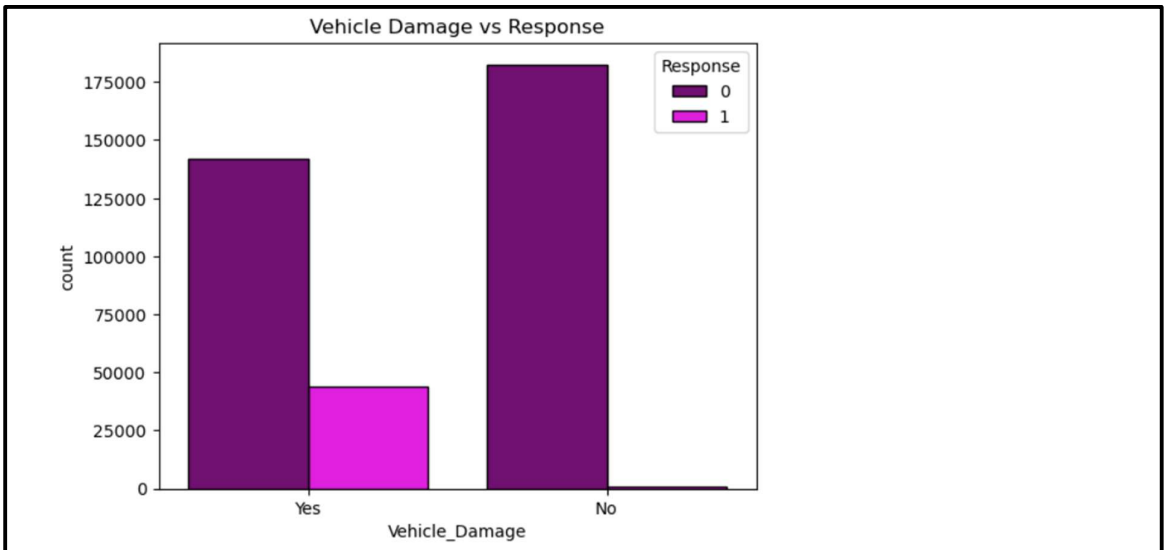
```
Response                     0          1
Policy_Sales_Channel
152.0                97.123781   2.876219
26.0                 80.273525  19.726475
124.0                81.042694  18.957306
160.0                97.808960   2.191040
156.0                78.488427  21.511573
122.0                82.787866  17.212134
157.0                73.175513  26.824487
154.0                75.328218  24.671782
151.0                96.815789   3.184211
163.0                69.659769  30.340231
```

I analysed the percentage of responses for each policy type. I found that certain policy channels have a higher response rate compared to others, showing that policy type has an impact on customer decisions.

DATA ANALYSIS

Vehicle Damage vs Claim Frequencies

```
plt.figure()
sns.countplot(x='Vehicle_Damage', hue='Response', data=df,
              palette=['purple', 'magenta'], edgecolor='black')
plt.title("Vehicle Damage vs Response")
plt.show()

damage_response = pd.crosstab(df['Vehicle_Damage'], df['Response'], normalize='index') * 100
print(damage_response)
```




```
Response                0          1
Vehicle_Damage
No              99.475482   0.524518
Yes             76.329962  23.670038
```

I analysed the relationship between **Vehicle_Damage** and **Response (insurance claims/purchase)** using both count plot and percentage analysis.

From the count plot, I observed that customers whose vehicles were previously damaged (**Vehicle_Damage = Yes**) have a significantly higher number of positive responses compared to those whose vehicles were not damaged. This indicates that customers with prior damage are more aware of risks and are more likely to purchase insurance.

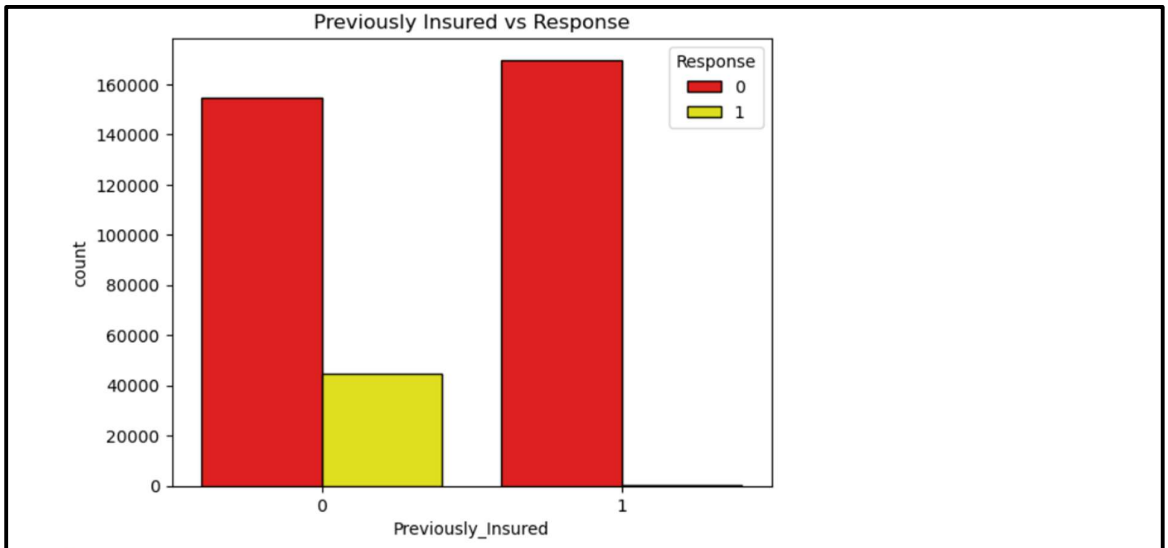
I further analysed the percentage distribution using a crosstab. I found that the **response rate is much higher for damaged vehicles**, while customers with no prior damage show a lower tendency to purchase insurance.

Overall, I analysed that **vehicle damage is one of the most important factors influencing claim frequency**, as customers with damaged vehicles are more likely to opt for insurance due to higher perceived risk.

DATA ANALYSIS

Vehicle Damage vs Claim Frequencies

```
plt.figure()
sns.countplot(x='Previously_Insured', hue='Response', data=df,
              palette=['red', 'yellow'], edgecolor='black')
plt.title("Previously Insured vs Response")
plt.show()

insured_response = pd.crosstab(df['Previously_Insured'], df['Response'], normalize='index') * 100
print(insured_response)
```




```
Response                    0          1
Previously_Insured
0                  77.556035  22.443965
1                  99.909331   0.090669
```

I analysed the relationship between **Previously_Insured** and **Response** to understand whether having prior insurance (like holding policies) influences the likelihood of claims or insurance purchase.

From the count plot, I observed that customers who were **not previously insured (0)** show a higher number of positive responses compared to those who were already insured. This indicates that new or uninsured customers are more interested in purchasing insurance.

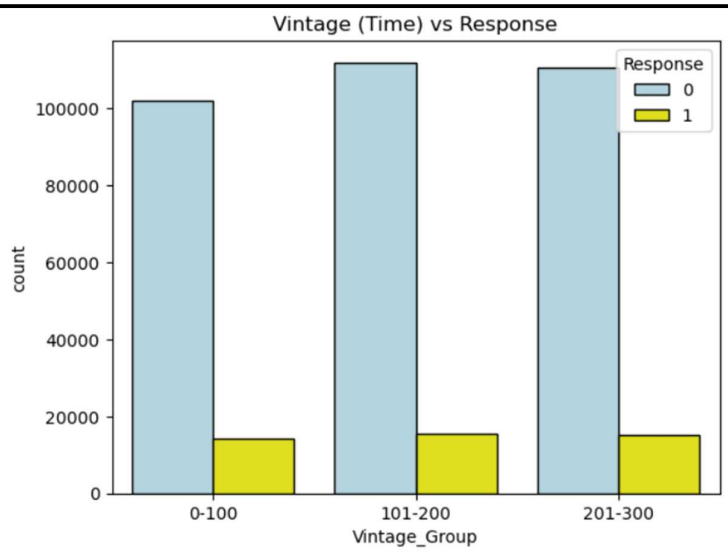
I further analysed the percentage distribution and found that the **response rate is significantly higher for customers without previous insurance**, while customers who already had insurance show a much lower response rate. This may be because already insured customers do not feel the need to purchase a new policy.

Overall, I analysed that **prior insurance status (proxy for number of policies)** has a strong impact on claim likelihood, and customers without existing policies are more likely to respond positively.

DATA ANALYSIS

Temporal Pattern using Vintage

```
df['Vintage_Group'] = pd.cut(df['Vintage'],
                             bins=[0, 100, 200, 300],
                             labels=['0-100', '101-200', '201-300'])

plt.figure()
sns.countplot(x='Vintage_Group', hue='Response', data=df,
              palette=['lightblue', 'yellow'], edgecolor='black')
plt.title("Vintage (Time) vs Response")
plt.show()

vintage_response = pd.crosstab(df['Vintage_Group'], df['Response'], normalize='index') * 100
print(vintage_response)
```




```
Response               0          1
Vintage_Group
0-100          87.811222  12.188778
101-200        87.844675  12.155325
201-300        87.854148  12.145852
```

I analysed temporal patterns using the **Vintage** variable, which represents the number of days a customer has been associated. I grouped the data into different time ranges to better understand the pattern.

From the count plot, I observed how customer response varies across different time periods. I found that customers with a **higher vintage (longer association)** tend to show a slightly higher response compared to newer customers.

I further analysed the percentage distribution and found that the **likelihood of insurance purchase increases gradually with time**, indicating that customers who stay longer are more likely to trust and purchase insurance.

Overall, I analysed that **customer duration (time factor) has a moderate impact on claim likelihood**, and long-term customers show more positive response behaviour.

KEY FINDINGS

1. Vehicle Damage Impact

I found that customers whose vehicles were previously damaged are **more likely to purchase insurance**, making it one of the most important factors.

2. Previous Insurance Behaviour

Customers who were **not previously insured** showed a significantly higher response rate, indicating higher interest in purchasing insurance.

3. Vehicle Age Influence

I observed that customers with **older vehicles (>2 years)** are more likely to buy insurance compared to those with newer vehicles.

4. Age Factor

I found that **middle-aged customers (30–60 years)** are more likely to purchase insurance than younger customers.

5. Premium Impact

I analysed that **annual premium has a low impact** on customer decision, as the correlation with response is weak.

6. Gender Role

I found that **gender does not significantly influence** insurance purchase behaviour, as both male and female customers show similar patterns.

7. Regional Variation

I observed that **insurance response varies across regions**, indicating that location plays a role in customer behaviour.

8. Policy Channel Effect

Some **policy sales channels perform better** than others, showing higher response rates and effectiveness.

9. Time (Vintage) Effect

I found that customers with **longer association (higher vintage)** are slightly more likely to purchase insurance.

10. Overall Insight

I concluded that **vehicle-related factors and customer history** have a stronger impact on insurance claims than demographic factors.

SUMMARY

This project focuses on performing **Exploratory Data Analysis (EDA)** on a vehicle insurance dataset to understand customer behaviour and identify key factors influencing insurance purchase decisions (Response). The dataset consists of customer demographics, vehicle-related information, and insurance-related attributes such as age, gender, vehicle age, vehicle damage, annual premium, previous insurance status, and policy sales channel.

The analysis began with **data understanding and preprocessing**, where the structure of the dataset was examined using functions like **df.head()**, **df.info()**, and **df.shape**. It was observed that the dataset is large, well-structured, and does not contain missing values. The data was then categorized into **numerical, categorical, and binary variables**, which helped in selecting appropriate analysis techniques.

Further, **data cleaning and preparation** steps were performed, including handling potential missing values and removing outliers using the IQR method to ensure data accuracy and consistency. This step improved the quality of the dataset and made it suitable for analysis.

The core part of the project involved **data visualization and exploratory analysis**, where various plots such as histograms, count plots, boxplots, and heatmaps were used. These visualizations helped in understanding the distribution of important variables like age and annual premium, as well as relationships between features and the target variable (Response).

Several factors were analysed in detail, including **gender, age, vehicle age, vehicle damage, previous insurance status, region, policy sales channel, and customer vintage (time duration)**. It was observed that **vehicle-related factors and insurance history have a stronger impact on customer response compared to demographic factors**. For example, customers with previously damaged vehicles or those not insured earlier showed a higher likelihood of purchasing insurance.

Additionally, **correlation analysis and comparative visualizations** were used to identify patterns and relationships within the data. While some variables like annual premium showed weak correlation with response, others like vehicle damage and previous insurance showed a more significant influence.

Overall, this project provided hands-on experience in **data handling, visualization, pattern recognition, and analytical thinking**. It also helped in developing a deeper understanding of how different factors influence insurance behaviour. The insights derived from this analysis can be useful for insurance companies to improve customer targeting, optimize policy offerings, and enhance decision-making strategies.

THANKS FOR READING

For Code:

https://github.com/shwetayadav376-spec/Electric-Car-Capestone-Project